

AI SWING & MOMENTUM TRADING SYSTEM

Blueprint V2.1

Autonomous Long/Short Swing Trading with Deterministic Risk + Selective AI Research

Updated August 24, 2026

End-state vision

Fully automated discovery, ranking, entry, position management, pyramiding, exits, short-covering, strategy allocation and portfolio de-risking. Humans govern models, capital, exceptions, broker authentication/session maintenance and the kill switch - not individual trades.

Research and engineering blueprint. Performance targets are hypotheses for validation, not guarantees. This document supersedes Revised V2 where V2.1 changes are stated.

V2.1 Change Summary

The V2 architecture remains valid. V2.1 tightens the production boundary around broker operations, event/gap risk, short-side mechanics, fill protection, validation and AI provenance.

Area	V2.1 decision
Broker autonomy	Human trade approval remains unnecessary, but broker-required authentication/session maintenance is an explicit operational responsibility.
Data stack	Use survivorship-bias-aware LEAN/QuantConnect data initially; treat exact earnings timestamps, analyst revisions and historical borrow conditions as Phase-0 blocking items.
Risk	Add gap/event-risk budgets; do not treat stop distance as total swing-trade risk.
Execution	Protect partial fills immediately; broker reconciliation must cover positions, cash and open orders.
Shorts	Require borrow availability/fee checks at signal and pre-submit, plus squeeze/SSR/recall controls.
Validation	Paper proves logic and operations; micro-live proves real fills, slippage, borrow and routing.
Allocator	Start with strategy risk caps/opportunity usage; enable performance-based dynamic allocation only after enough evidence.
AI	Every AI output requires timestamped source provenance, model version and prompt version; AI influence remains bounded.
V1 scope	Keep X/social and aggregate options-positioning features out of V1.

1. Executive Summary

V2.1 preserves the simpler autonomous design of Revised V2. Breakout remains a first-class core strategy alongside Momentum Pullback and Post-Earnings Announcement Drift (PEAD). AI is deliberately constrained to qualitative information processing and challenge analysis; scanning, risk, order approval and money management remain deterministic.

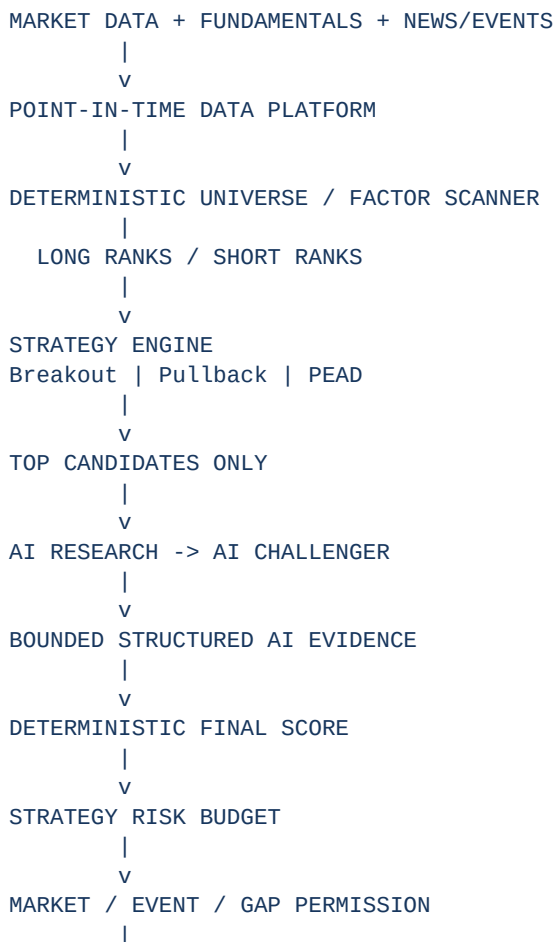
Locked principle

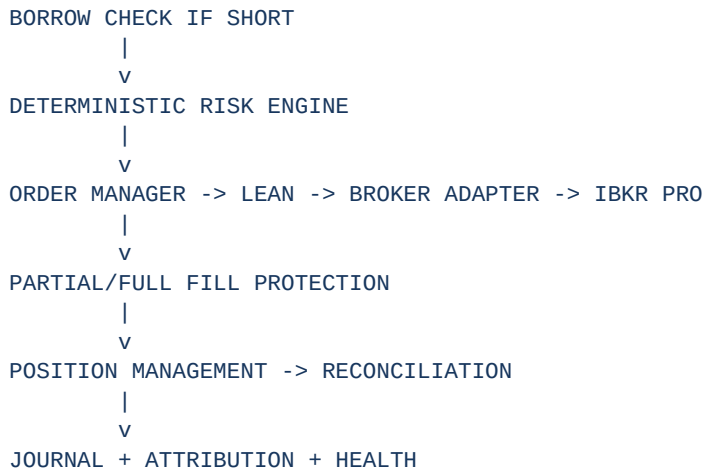
AI may improve information processing. Mathematics and deterministic software control money, risk and broker actions.

2. Mission and Operating Model

- Trade liquid U.S. equities long and short with a primary 2-30 trading-day horizon.
- Automatically discover, rank, enter, pyramid, monitor and exit positions.
- Use breakout, pullback and earnings-driven momentum as independent alpha engines.
- Allocate capital among strategies using controlled risk budgets and, later, measured strategy health.
- Minimize human intervention after live validation; retain human governance, broker-required authentication/session maintenance and an independent emergency kill switch.
- Keep the broker replaceable through a broker-adaptor interface.

3. End-to-End Architecture





4. Tradable Universe

Filter	Initial V2.1
Security	NYSE/NASDAQ common stocks
Price	>\$10
Market cap	>~\$1.5B
Average daily dollar volume	>~\$25M
History	>250 trading days
Initial exclusions	OTC, warrants, preferreds, leveraged ETFs, microcaps, extremely illiquid biotech, binary special situations, extreme HTB/squeeze names

5. Core Strategy Engines

5.1 Momentum Breakout - Core

Breakout is not treated as 'new high = buy.' The system looks for a strong trend, compact base or resistance structure, confirming relative strength, volume expansion, acceptable fundamental/revision context and market permission.

- Price > 50 DMA; 50 DMA > 200 DMA
- Relative strength >= ~80-85th percentile
- Positive sector-relative / residual momentum
- 20-60 day consolidation or well-defined resistance
- Breakout confirmation
- Volume >= ~1.5x baseline
- Revisions neutral-to-positive
- Adequate liquidity
- Long exposure permitted by trend/volatility state

Short breakdown logic is separate and stricter: relative weakness, support failure, negative revisions/fundamentals, selling volume, weak sector, acceptable borrow and limited squeeze risk.

5.2 Momentum Pullback - Core

Continuation template

Established trend + high relative strength + favorable sector context + controlled pullback toward 10/20 EMA or structure + contracting pullback volume + renewed demand/reversal confirmation = candidate continuation entry.

5.3 PEAD / Earnings Momentum - Core

Post-event evidence

EPS surprise + revenue surprise + guidance change + post-earnings estimate revisions + abnormal volume + price gap/repricing + stabilization or continuation confirmation = long or selective short candidate.

The system trades the post-announcement information path rather than attempting to predict the earnings result. By default, Breakout and Pullback positions do not carry through earnings unless an explicitly tested event-risk rule permits it.

5.4 Peer Catalyst Momentum - Phase 2

Use AI to map material earnings or industry information from one company to suppliers, peers, competitors and customers; then trade only peers that also pass quantitative confirmation.

6. Ranking and Factor Architecture

V2.1 retains a small number of cross-sectional rank/Z-score composites rather than a large manually weighted checklist.

Composite	Initial weight	Examples
Earnings / Revision Momentum	~35-40%	EPS/revenue surprise, guidance, revision velocity, margin acceleration
Relative / Residual Momentum	~35-40%	1/3/6m momentum, SPY-relative, sector-relative, residual momentum
Price / Volume Quality	~20-30%	abnormal volume, volatility contraction, breakout/pullback quality, trend persistence

Machine-learned weights are a later experiment, not a starting dependency. Simple factors must first remain robust out of sample.

7. AI Scope - Deliberately Narrow

7.1 No AI Scanner

Python/SQL scans the full universe and computes quantitative ranks. AI does not inspect ~1,200 tickers to decide which ones 'feel interesting.'

7.2 Research Agent

Receives only top-ranked candidates and extracts structured evidence from earnings releases, transcripts, 8-K/10-Q filings, guidance, contracts, products, regulatory events and relevant industry news.

ticker
as_of_time
source_ids
source_publish_times
catalyst_type
guidance_change
demand_change
margin_change
competitive_position
management_tone
risk_flags
catalyst_score
confidence
model_version
prompt_version

7.3 Challenger Agent

Attempts to falsify the trade thesis and surface contradictions, crowding, valuation, event risk and alternative explanations.

7.4 Production Boundary

No AI risk engine / no arbitrary AI trading

AI cannot choose dollar size, override risk limits or send arbitrary orders to the broker. AI may contribute bounded structured evidence. Deterministic code approves and sizes a trade; the execution engine sends the approved order automatically.

8. Strategy Capital Allocation

Do not begin live V1 with a fast performance-chasing allocator. Start with per-strategy risk caps plus opportunity-based usage. After each engine has adequate forward evidence, enable a slow-moving allocator using trailing expectancy, drawdown, rolling risk-adjusted performance, current opportunity count, realized volatility and cross-strategy correlation.

Strategy	Initial risk-budget cap (illustrative)	Later allocator input
Breakout	<=45% of active strategy risk	trend quality, expectancy, opportunity count
Pullback	<=45%	continuation quality, expectancy, volatility
PEAD	<=35%	event opportunity count, post-gap expectancy

Caps are not target weights and do not need to sum to 100%. Unused risk remains unallocated.

9. Return-Enhancement Mechanisms

- Residual and sector-relative momentum to isolate stock-specific strength.
- Earnings-revision velocity rather than static analyst ratings.
- Pyramiding winners only after confirmation while ratcheting stops higher.
- Allowing large winners to run with volatility/structure-aware trailing exits.
- Peer-catalyst momentum for information spillovers (Phase 2).
- Dynamic strategy allocation only after minimum evidence thresholds.

- Selective shorts rather than forced market neutrality.
- Execution optimization by strategy: confirmation, VWAP, opening-range behavior, patient limits and post-gap stabilization.
- Volatility targeting to improve capital efficiency and later introduce measured leverage.
- Momentum-crash protection during violent factor reversals.

10. Deterministic Risk and Position Management

Control	Initial V2.1
Long planned risk/trade	0.50% of equity
Short planned risk/trade	0.25% of equity
Maximum open planned risk	~5%
Maximum individual position	~8-10%
Initial gross short exposure	<=25%
Portfolio volatility target	~10-12% annualized
Leverage	None initially
Averaging down	Disabled
Pyramiding	Winners only; confirmation required

10.1 Position Sizing

Base sizing

risk_dollars = portfolio_equity x allowed_risk

risk_per_share = abs(entry - technical_stop)

base_shares = risk_dollars / risk_per_share

Then cap by maximum position %, sector/correlation exposure, portfolio volatility target, liquidity and borrow/short constraints.

10.2 Gap / Event Risk Budget

- Technical stop risk is not assumed to equal total loss risk.
- Every candidate carries an overnight/event-risk estimate and upcoming-event flag.
- Breakout/Pullback positions do not carry through earnings by default.
- Portfolio-level correlated gap exposure is capped.
- PEAD enters after the event, not before it.

10.3 Pyramiding Winners

Initial entry ~0.30% planned risk -> first confirmation add ~0.15% -> second confirmation add ~0.10%. At every add: move stop/reduce original open risk, re-check sector/portfolio correlation and never add automatically to a losing trade.

11. Market Permission and Volatility Scaling

- Broad-market long-term trend/breadth determines how much directional risk is permitted.
- Realized market and portfolio volatility continuously scale gross exposure.
- High-volatility/reversal environments reduce new exposure, pyramiding and short aggression.
- Low/moderate volatility permits normal exposure; leverage remains a future stage only after live validation.

12. Short-Side Architecture

Short alpha is a separate strategy family, not the inverse of the long book.

- Borrow availability required at signal time and re-checked immediately before submission.
- Borrow fee must remain below a strategy-specific threshold.
- Track shortable quantity, fee deterioration and borrow availability changes.
- Hard-to-borrow and squeeze-prone names excluded initially.
- SSR/short-sale restriction state is checked before execution.
- Binary events checked before carrying a short.
- Recall/buy-in and corporate-action handling are explicit operational states.
- Short risk is half the initial long risk.
- Gross short exposure capped initially at ~25%.
- Historical short backtests are discounted unless borrow cost and availability are modeled conservatively.

13. Portfolio Circuit Breakers

Trigger	Action
Daily portfolio loss about -2%	Stop initiating new risk; do not blindly liquidate all positions
Weekly loss about -4%	Reduce risk on new trades
Portfolio drawdown about -8%	Reduce gross exposure / risk tier
Portfolio drawdown about -12%	Stop new entries and trigger diagnostic review
Broker/data-integrity failure	Stop new orders; preserve deterministic exits where safe
AI outage	Continue deterministic position management; block AI-dependent new entries

14. Autonomous Trading Lifecycle

SCAN -> STRATEGY MATCH -> AI RESEARCH -> CHALLENGER -> FINAL STRUCTURED SCORE -> STRATEGY RISK BUDGET -> MARKET/EVENT/GAP PERMISSION -> RISK + BORROW CHECK -> EXECUTION LOGIC -> ORDER -> PARTIAL/FULL FILL EVENT -> IMMEDIATE PROTECTION FOR FILLED QUANTITY -> PYRAMID / TRAIL / EXIT -> BROKER RECONCILIATION -> JOURNAL -> ATTRIBUTION -> STRATEGY HEALTH

In mature production, normal trades complete this lifecycle without human trade approval.

15. Human Control Plane

- Global auto-trading ON/OFF.

- Disable new entries while preserving exits.
- Long-only / short-disable modes.
- Disable a strategy or ticker.
- Reduce risk multiplier.
- Cancel all unfilled orders.
- Emergency flatten.
- Approve model/parameter changes and capital scaling.
- Review system-health exceptions and reconciliation failures.
- Perform broker-required authentication/session maintenance.

Human role

Govern the system, not individual trades. The kill switch must be independent of the AI. Broker-required authentication/session maintenance is an operational exception, not trade approval.

16. Autonomous-Operations Requirements

- Deterministic client/order IDs for idempotency and duplicate-order prevention.
- Frequent reconciliation of broker vs internal positions, cash and open orders.
- Freshness guards for price, news and feature data.
- Broker disconnect detection, reconnect and state reconciliation before resuming.
- Separate research, paper and live environments.
- Secrets manager for brokerage/provider credentials.
- Strategy version + code commit recorded with every trade.
- External alerts for broker, data, database, order and risk anomalies.
- Independent emergency stop/flatten path.
- Second execution username/session before live automation to avoid accidental manual-session displacement.

17. Recommended V2.1 Tool Stack

Layer	V2.1 choice	Role
Research/backtest	QuantConnect LEAN	Backtests, paper/live algorithm engine, portfolio/order simulation
Broker	Interactive Brokers Pro	Primary long/short execution behind adapter
Historical price/security master	LEAN/QuantConnect survivorship-bias-aware U.S. equity data initially	OHLCV, delistings, symbol changes, corporate actions
Live scan data	QuantConnect SIP or audited consolidated feed	Swing-resolution price, volume, quote data
Earnings/revisions	TBD after point-in-time audit	Exact event times, consensus, revision history, guidance
Borrow data	IBKR live + evaluated historical provider	Shortability/fees; conservative historical modeling

Layer	V2.1 choice	Role
News/events	Structured provider TBD	Earnings, analyst and event data
AI	LLM API with structured outputs	Research + challenger only
Database	PostgreSQL / optional TimescaleDB	Features, signals, trades, audit state
Runtime	Python + Docker	Services and reproducible deployment
Dashboard	Web command center	Health, exposures, positions, controls, kill switch
Source control	GitHub	Versioning, CI/CD and auditable deployments

18. Phase 0 - Point-in-Time Data Feasibility

- Historical prices including delisted names and symbol changes.
- Splits/dividends/corporate actions.
- Exact earnings publication timestamps.
- Point-in-time analyst consensus and revision history.
- Guidance history.
- Historical news/event timestamps.
- Historical short borrow fees and availability, or a deliberately conservative substitute model.

Do not skip

The hardest data problems are point-in-time analyst revisions and realistic historical short borrow conditions - not ordinary OHLCV.

19. Backtesting and Research Methodology

- Test Breakout Long, Breakdown Short, Pullback Long, PEAD Long and PEAD Short independently.
- Then test combined portfolio, strategy risk budgets/allocator, volatility targeting, pyramiding and execution variants.
- Use multiple market environments, ideally approximately 2010-2026 where point-in-time data permits.
- Include commissions, spread/slippage, realistic fills, short borrow cost and conservative liquidity assumptions.
- Control survivorship bias and future/revised fundamental data.
- Use walk-forward and locked out-of-sample periods.
- Only after basic edge survives, add parameter-stability tests, multiple-testing controls and stronger statistical validation.

19.1 AI Incremental-Alpha Test

A/B/C test

Model A: deterministic strategy only

Model B: A + structured earnings/news/event inputs

Model C: B + LLM qualitative interpretation

Keep AI in the trade-ranking path only if C materially improves out-of-sample economics and risk-adjusted

performance over B.

20. Success Metrics

Metric	Planning range	Interpretation
Net annualized return	12-18% acceptable; 18-25% strong; 25%+ exceptional	Do not optimize directly to a target number
Maximum drawdown	Prefer <20-25%	Capital survival constraint
Sharpe	>~1.0 useful; >1.2 strong	Out-of-sample after costs
Profit factor	>~1.4	Trade economics
Live slippage	Near modeled assumptions	Operational reality
Strategy health	Positive expectancy independently	Avoid dependence on one engine

21. Build Roadmap and Realistic Timeline

Phase	Primary deliverable	Typical window	Capital mode
0. Data feasibility	PIT provider audit, schemas, sample datasets, short-data feasibility	1-2 weeks	None
1. Data + factor foundation	Ingestion, universe, features, reproducible historical snapshots	2-3 weeks	None
2. Breakout engine	Long breakout + short breakdown research/backtests	2-3 weeks	None
3. Pullback + PEAD	Independent modules and event infrastructure	2-4 weeks	None
4. Portfolio/risk	Risk budgets, correlation, vol target, gap risk, stops, pyramiding, breakers	2-3 weeks	None
5. Broker + automation	LEAN/IBKR adapter, orders, idempotency, reconciliation, health controls	2-3 weeks	Paper
6. AI layer	Research/challenger structured outputs and A/B/C evaluation	1-2 weeks	Paper
7. Operational hardening	Dashboard, kill switch, alerts, restart/recovery and audits	1-2 weeks	Paper

Phase	Primary deliverable	Typical window	Capital mode
8. Forward paper run	Automated paper trading and operational validation	4-8+ weeks	Paper
9. Micro-live	\$10K-\$25K class autonomous deployment, no leverage	4-8+ weeks observation	Small live

Practical planning estimate

A credible automated V1 codebase can be built in roughly 10-16 weeks if data access is resolved quickly. A version trusted with meaningful autonomous capital is more realistically a 4-6 month program because forward validation and operational hardening cannot be compressed safely.

22. Deployment Gates

Mode	Behavior	Exit gate
Research	No orders	Strategies survive realistic out-of-sample tests
Fully automated paper	No human trade approvals	Operational state stable; no duplicate/missing orders; risk/reconciliation/restart logic stable
Micro-live autonomous	Small capital; no leverage	Real fills, slippage, borrow, routing, risk and strategy behavior track expectations
Scaled autonomous	Progressively larger capital	Repeated statistical and operational review
Controlled leverage	Volatility-targeted incremental leverage only	Long live record with controlled drawdown and reliable infrastructure

Paper validates logic and operations. Micro-live validates actual fills, partial fills, slippage, borrow availability/fees and routing.

23. Future Alpha / Return Expansion

- Peer-catalyst momentum.
- Sector-neutral relative-strength portfolios.
- Advanced earnings-revision models.
- Alternative data only if it proves incremental alpha.
- Strategy-specific execution optimization.
- Options/index hedges where they improve risk-adjusted return.
- Volatility-targeted controlled leverage after live proof.
- Capacity-aware universe expansion as capital grows.
- Social/X sentiment and options-positioning features only after core V1 edge is proven.

24. Locked V2.1 Specification

Decision	Locked V2.1
Universe	~1,200 liquid U.S. equities

Decision	Locked V2.1
Direction	Long + short
Core strategies	Momentum Breakout + Momentum Pullback + PEAD
Phase-2 strategy	Peer Catalyst Momentum
Ranking	Earnings/revisions + relative/residual momentum + price/volume quality
Scanner	Deterministic Python/SQL
AI	Research Agent + Challenger Agent; bounded structured evidence with provenance
Risk	Deterministic + gap/event-risk layer
Trade execution	Automatic after deterministic approval
Human trade approval	Not required in mature production
Broker	IBKR Pro behind adapter
Engine	QuantConnect LEAN initially
Database	PostgreSQL
Long risk	0.50% planned risk/trade
Short risk	0.25%
Max open planned risk	~5%
Initial short gross	<=25%
Initial leverage	None
Volatility target	~10-12%
Pyramiding	Yes - winners only
Social/X	Not V1
Options positioning	Not V1
Human role	Governance, model/capital changes, exceptions, broker authentication/session maintenance, kill switch

25. IBKR Account Configuration Baseline

Current account status observed from the supplied settings screenshot (sensitive identifiers intentionally omitted): Individual account, Margin account type, IBKR Pro pricing, USD base currency, paper trading account present, Read-Only Access enabled, and Stock Yield Enhancement Program enabled.

Setting	Recommended V1 state	Reason
Pricing plan	IBKR Pro	Required/appropriate for automated API integration and project use
Account type	Margin / Reg T	Needed for short selling; no need for Portfolio Margin at ~\$80K
Base currency	USD	Matches U.S. equity strategy
Paper trading	Enabled	Primary development/forward-validation environment
Trading permissions	U.S. Stocks	V1 trades liquid U.S. common stocks only
Options/Futures permissions	Off initially	Reduce operational surface area
Leverage	Do not use initially	Blueprint remains no-leverage V1
Read-Only Access	May stay enabled	It only provides a quick read-only mode; full login is still required to trade
Secure login	IBKR Mobile/IB Key or passkey	2FA required for secure operation
Stock Yield Enhancement	Prefer OFF during V1 live testing (optional)	Simplifies tax/corporate-action/reconciliation behavior; not required for our strategy
Market data	Streaming U.S. equity data for manual verification; scanner data remains independent	Avoid broker-data coupling
Second username/session	Create before live automation	Prevents manual login from displacing the execution session

Portfolio Margin

Do not upgrade now. Current IBKR documentation requires at least ~\$110K NLV to upgrade an existing account and maintains a \$100K minimum threshold; our initial ~\$80K capital does not justify it.

26. IBKR/LEAN Operational Notes

- Design live deployment around IBKR Pro with a dedicated execution username/session before automation.
- Keep a separate manual/control-plane login where possible.
- Treat broker authentication/session maintenance as a scheduled operational responsibility, not an AI task.
- Use broker account state as the execution source of truth; reconcile frequently against internal state.

- Do not use broker market data as the sole source for universe ranking/backtests.
- Keep the same broker-adaptor abstraction so IBKR can be replaced or supplemented later.

27. End-State Vision

Autonomous system

Discover -> Analyze -> Rank -> Challenge -> Allocate -> Risk -> Trade -> Manage -> Pyramid -> Exit -> Reconcile -> Measure -> Learn

Human oversight remains above the system: strategy governance, capital scaling, parameter/model releases, exception handling, broker-required authentication/session maintenance and an independent kill switch.

Final design objective

A quantitative autonomous trading platform that uses AI where it demonstrably improves information processing, while deterministic software controls money, risk and broker actions.

Appendix - Current Tool Verification (August 2026)

QuantConnect LEAN - Interactive Brokers live trading

<https://www.quantconnect.com/docs/v2/lean-cli/live-trading/brokerages/interactive-brokers>

Interactive Brokers API documentation

<https://www.interactivebrokers.com/campus/ibkr-api-page/ibkr-api-home/>

IBKR Portfolio Margin eligibility

<https://www.interactivebrokers.com/en/trading/marginRequirements/marginPortfolio.php>

IBKR Market Data Pricing

<https://www.interactivebrokers.com/en/pricing/market-data-pricing.php>

IBKR Secure Login System

<https://www.interactivebrokers.com/en/general/secure-login.php>

IBKR Read-Only Access guide

<https://www.ibkrguides.com/orgportal/usersettings/readonlyaccess.htm>

Tool capabilities and fees can change. Re-verify official documentation immediately before implementation and before enabling live automation.